

Which literature belongs beside our testbed?

A coaxial, counter-rotating propulsion unit on a two-axis gimbal. Compare demonstrated control architectures without mixing different actuation capabilities.

Electric TVC: controller architecture × experimental platform

Seven selected studies | first-author labels | no color, shape or size encoding

Coaxial-gimbal vehicle (close actuation topology)	• Denton (2022) Attitude feedback; launch-to-hover • Chen (2024) Nonlinear allocation + damping • Santos (2026, E-Rocket) PID trajectory-tracking baseline	• Spannagl (2021) Optimal G; MPC + inner feedback • Linsen (2022) Optimal G; joint position/attitude NMPC		
Articulated rocket surrogate (different thrust actuator)			• Santos (2026, QuadRocket) Adaptive backstepping + actuator dynamics	
Constrained TVC apparatus (one-axis experiment)				• Osedo (2023) PPO + domain randomization; pitch only
	Feedback + explicit allocation	Optimization-based control / G+C	Adaptive nonlinear control	Direct reinforcement learning
Control architecture (categorical; no ranking)				

Within-cell spacing separates labels only. Empty cells are not evidence of novelty. Detailed GNC roles and validation limits are in the companion matrix.

Scope: the six platform papers from the review + Osedo's adjacent one-axis experiment. Denton (2022) is the single platform representative; Denton (2025) remains an identification follow-up.

Read the empty cells carefully: this is a categorical literature-positioning map, not a performance ranking or a proof of novelty. The separate transfer-method section retains Li, Torrente and Zhang without treating their vehicles as ours.

Compare the control stack, not only the algorithm name

G = guidance; N = navigation/state and disturbance estimation; C = tracking/stabilization; A = allocation/actuation.

Paper / source	G	N	C + A	Evidence / caveat
Spannagl (2021)	Optimal free-time guidance; retargeting	State EKF + offset filter	Offset-free MPC + inner feedback; inverse allocation; thrust lag modeled	Indoor + 19 outdoor missions; onboard G/C
Linsen (2022)	Optimal free-time terminal-target guidance	Pixhawk filter + augmented EKF	NMPC jointly tracks position and attitude; servo/motor constraints	Indoor tracking/wind; outdoor apogee + landing
Santos (2026, E-Rocket)	Offline trajectory; mission logic	PX4 EKF + indoor motion capture	PID cascade + allocation; simplified actuation model	Indoor flights; IFAC accepted/program-listed
Santos (2026, QuadRocket)	Supplied reference trajectory	Motion-capture-supported feedback	Adaptive backstepping + dynamic-surface control; different actuator	Flight experiments; main tracking loop offboard
Chen (2024)	Ascent target; manual maneuvers also shown	IMU/ToF; separate Pixhawk hover	Nonlinear allocation + damping; two serial tilt servos	Custom ascent and Pixhawk hover are different stacks
Denton (2022)	Pilot commands; launch-to-hover	Onboard attitude estimation	Cascaded feedback; gimbal + differential rotor speed	Rig + free flight; not autonomous landing guidance
Osedo (2023)	Prescribed pitch target	Pitch/rate feedback	PPO/LSTM + domain randomization; thrust/vectoring increments	Pitch-only rig; out-of-range model case fails

Correction from the final Spannagl PDF: 100 m maximum altitude, 50 m maximum diversion and 19 successful outdoor flights are supported (Sec. IV-C, p.6350). First-order thrust response is already in its MPC model (Eq.2e).

Research attribution: hold G and N fixed when testing C. Existing state estimation is not, by itself, a new navigation contribution.

Keep the useful mechanism; preserve the plant boundary

These are supporting method references, not extra points in the seven-paper main map.

Reference	Mechanism worth transferring	Boundary for our project
<u>Torrente (2021)</u> Residual-model MPC	Learn dynamics error with Gaussian processes; insert it into MPC. Compare nominal, simple fitted and learned corrections.	Fixed-rotor quadrotor, not TVC. Supervised learning, not RL. Benefits depend on operating regime; low-speed improvement is not guaranteed.
<u>Li (2024)</u> Servo-integrated NMPC	Include actuator response in prediction/control; test the effect of the actuator model.	Four tilttable rotors are overactuated. Servo ablation in simulation must not be presented as a hardware ablation.
<u>Zhang (2026)</u> Direct actuator-level RL	PPO + actuator identification + latency alignment + domain randomization; same-platform comparison against Li-style NMPC.	This is the uploaded PDF, not Cuniato (2024). Four rotor/servo pairs. RA-L acceptance reported; final publication metadata unverified.

Zhang's hardware results show a trade-off, not an overall RL victory

Waypoint tests: mean steady-state position-error norm 0.077 m (RL) vs 0.042 m (NMPC); orientation-error norm 4.261 vs 4.350 degrees (Table V, three sequence repeats). Trajectory tests favor NMPC in mean position and orientation error (Sec. III-D, five repeats). Allocation-singularity traversal is simulation only; the 90-degree hardware case uses a reconfigured plant and retrained policy.

Keep separate: residual-model learning changes the prediction model; residual RL adds a learned action correction to a baseline controller. Neural-Fly adapts coefficients of learned features and is not RL. Neural-Fly and Liu remain idea notes, not main-map points.

Most useful lesson from Torrente: test against a simple identified correction, not only an uncorrected nominal model. A learned method must earn its additional data and computation cost on our own platform.

Choose a measurable limitation before choosing a controller

A proposed sequence for discussion, not a claim that the listed questions are already novel.

Candidate question	Prior-art boundary	Smallest decisive experiment
A. Actuator fidelity When does measured motor/gimbal response improve constrained tracking?	Spannagl models thrust lag; Linsen constrains actuator rates; Li integrates servo dynamics.	Static vs identified dynamic actuator model, with the same control architecture, estimator, reference and limits.
B. Residual-model value What remains after physical identification and simple compensation?	Offset estimation and learned residual models already exist. A new platform alone is insufficient.	Identified baseline vs simple correction vs learned residual. Use shared training data and held-out complete runs.
C. Learned-control trade-off What robustness/computation benefit survives a matched comparison?	Osedo provides constrained TVC RL; Zhang compares RL/NMPC on an overactuated plant.	One tuned conventional baseline vs one learned architecture. Match sensing/action authority; report failures and runtime.
D. Guidance feasibility Does a specific actuator-feasibility constraint improve tracking/landing?	Optimal guidance already exists in Spannagl and Linsen; identify the exact missing condition.	Hold tracking and estimation fixed; vary only guidance feasibility treatment. Requires reliable tracking first.

Immediate gate: safely measure actuator response, latency and static maps; validate on held-out runs; establish a tuned conventional baseline. Then choose A/B/C/D from the dominant measured limitation. This step does not commit the thesis to MPC or RL.

Minimum evidence: repeatable tracking error, peak error, saturation time, failure count and runtime distribution. Balance test order and battery state; fix G/N when evaluating C. Learning also needs data/compute and out-of-distribution reporting.

Scope/access record: Chih excluded; Elke retained only for surrogate-fidelity background; Denton (2025) not double-counted. The complete Markdown notes contain source locators, reading depth and limitations. No broad literature-completeness or novelty claim is made.